

DESIGN AND FABRICATION OF A BOMB DISPOSAL ROBOT

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Motivation

In high-risk environments such as military operations, counter-terrorism, and hazardous material handling, the safe disposal of explosive devices is a critical challenge. Traditional bomb disposal methods put human operators at significant risk. To mitigate these dangers, a **Bomb Disposal Robot** is proposed to enhance the safety and efficiency of explosive ordnance disposal (EOD).





Objective

1. Remote Handling of Explosive Devices

- **Safety of the Operator:** The robot allows a human operator to remain at a safe distance from the bomb while still interacting with it in real time.
- **Precision and Control:** The robotic arm can carefully manipulate components of the bomb, such as wires or buttons, to disable the device without triggering it.

2. Color Recognition for Identification of Critical Wires/Components

- **Wire Identification:** Many bombs use colored wires to distinguish critical connections (e.g., red for power, blue for a trigger, or green for a cut wire). The robot uses color sensors (such as the TCS3200) to identify the color of wires or components.
- **Prevent Human Error:** With color recognition, the robot can accurately differentiate wires and determine which ones are safe to cut or handle, preventing mistakes that could lead to triggering the device.
- **Autonomous Decision Making:** Based on the color of the wires, the robot can autonomously recommend or perform specific actions (e.g., cutting the red wire).

Objective

3. Real-Time Video Surveillance and Navigation

- **Camera Integration:** The robot can use cameras (possibly live-feed cameras and infrared cameras) to survey the bomb's surroundings and help the operator navigate the robot through obstacles.
- **Precise Movement:** The robot should be able to move around the bomb, positioning the robotic arm with high precision to interact with the device without making contact with other components of the bomb.

4. Risk Mitigation and Improved Efficiency

- **Reduced Risk to Human Life:** The robot handles the dangerous task of bomb disposal, reducing the potential risk to human lives during disarmament operations.
- **Faster and More Efficient:** With color recognition and real-time feedback, the robot can quickly make decisions and disarm the bomb more efficiently, reducing the time spent in hazardous environments.

Objective

5. Versatility and Adaptability

- **Handling Different Bomb Types:** The robot is designed to interact with various types of bombs, each with its unique wiring and components. The color recognition system helps the robot adapt to these variations.
- **Integration with Other Sensors:** The robot could also be equipped with temperature sensors, motion detectors, or even gas sensors to detect other potential hazards around the bomb.

6. Continuous Monitoring and Feedback

- **Human-Operator Interface:** The operator should receive continuous feedback (through a live video feed, sensor readings, and color detection status) to make informed decisions about disarming or handling the bomb.
- **Post-Disposal Reports:** The robot could document the bomb disposal process and share data (e.g., color of wires, steps taken) for training or later analysis.

Abstract

The development of a ***bomb disposal robot*** is a critical advancement in ensuring the safety and efficiency of bomb disposal operations. The primary objective of this project is to design a robotic system capable of identifying, interacting with, and disarming explosive devices remotely, with a focus on minimizing human risk. The robot uses a **4-DOF** robotic arm for precise manipulation of the bomb's components. The robot's movement and interaction with the bomb are controlled via a microcontroller, such as an **Uno max** , with real-time feedback provided through **video surveillance** to guide the human operator. This system allows for remote operation of the robot, keeping the operator at a safe distance while ensuring that **critical actions are performed with precision**. The integration of this technology is expected to improve safety, speed, and efficiency in bomb disposal scenarios, offering a reliable solution for handling potentially lethal devices. This project aims to create a more **autonomous, accurate, and safer method of bomb disposal**, contributing to the **reduction of casualties and the preservation of human life in hazardous situations**.

Introduction

- In recent years, the risk of explosive devices, particularly ***Improvised Explosive Devices (IEDs)***, has posed a significant threat to public safety, law enforcement, and military personnel.
- The core objective of this project is to design and develop a bomb disposal robot equipped with a color recognition system for wire identification, integrated with a ***4-DOF*** robotic arm to safely manipulate and disarm bombs.
- The robot is built around an ***Arduino-based control system*** for precise movement and handling of critical tasks. The ***4-DOF*** robotic arm allows for fine motor control, ensuring that even the most delicate tasks, such as cutting wires or removing components, can be performed with high accuracy.
- ***Real-time feedback***, including video, is transmitted to the human operator, who can make decisions based on the robot's findings.

Components Used

1. AURDINO UNO
2. L298N-MOTOR DRIVER
3. HC-05BLUTETOOTH MODULE
4. BREADBOARD
5. 8AA-BATTERIES
6. 4AA-BATTERIES CASE
7. JUMPERS
8. RED & WHITE LEDS
9. 100 OHMS RESISTORS
10. TRACKED CHASSIS
11. 4DOF-ROBOTIC ARM
12. POWER BANK
13. IP CAMERA

Working Principle

Step 1: Initialization and System Activation

When the robot is powered on, the microcontroller initializes the robotic arm, and communication systems. The operator can control the robot remotely through a user interface.

Step 2: Navigation to the Target

The operator navigates the robot using the remote interface. The chassis moves over different terrains, guided by the onboard camera for real-time monitoring.

Working Principle

Step 3: Object Handling

Once an object is detected: The robotic arm is positioned near the object. Using the 4-DOF control, the arm carefully grasps the object. The object is either relocated to a safe area or handled according to the disposal protocol.

Step 4: Safety Measures and Feedback

Safety Distance: The operator maintains a safe distance during operation.

Real-Time Feedback: Live video and sensor data are transmitted to the operator.

Emergency Stop: An emergency shutdown system is available for immediate deactivation.

Step 5: Object Disposal

Once the hazardous object is secured, the robot transports it to a disposal site. The robotic arm releases the object safely, ensuring minimal human intervention.

Existing Technology

Bomb disposal robots in use by the USA

MarkV-A1: Part of the Remotec ANDROS line, this robot is used by first responders to handle explosives and hazardous materials. In 2016, the Dallas Police Department used the MarkV-A1 to kill a gunman who had murdered five police officers.

Remotec Andros MarkV-A1

A bomb disposal robot with a 15-inch LCD screen, multiple cameras, microphones, lights, and sensors. It can be operated using a fiber optic cable, wireless radio control, or a hard tether cable.



Existing Technology

Bomb disposal robots in use by the USA

TALON: A medium-sized tactical robot that is used for explosive ordnance disposal (EOD). This family of robots is used by the military, law enforcement, and first responders. TALON robots have been used in Iraq, Afghanistan, and at Ground Zero after the 2001 World Trade Center attacks.

PackBot: This small unmanned ground vehicle (SUGV) is widely used by the US Army.



Existing Technology

Bomb disposal robots by the UK

In a groundbreaking development for military operations, the UK Ministry of Defense has successfully tested robot dogs in bomb disposal scenarios. The trial aimed at reducing the risk to military personnel by utilizing autonomous robots for dangerous tasks typically carried out by bomb disposal teams.

The four-day trial, conducted by the Defense Science and Technology Laboratory (Dstl). These scenarios included navigating stairs, autonomously opening and closing doors, and, most crucially, identifying and neutralizing explosive threats such as improvised explosive devices (IEDs). The robot dogs used sensors to classify potential threats before deploying disruptors to safely detonate or disarm bombs, significantly reducing human exposure to danger.



Bomb disposal robots



Limitations

Limited Degrees of Freedom (DOF) Challenge : With only 4 DOF, the arm lacks full flexibility compared to 6-DOF or 7-DOF robotic arms.

Impact: Some complex bomb disposal tasks requiring precise angling and wrist positioning may be difficult.

Possible Solution: Adding an extra wrist pitch or gripper roll DOF can improve dexterity.

Limited Payload Capacity Challenge: The robotic arm may struggle with heavy or large objects, depending on motor strength.

Impact: If an explosive device is too large or heavy, the arm may not be able to lift or manipulate it effectively.

Possible Solution: Use stronger actuators or hydraulic systems to increase lifting power.

Limitations

Difficulty in Handling Soft or Fragile Objects

Challenge: The gripper may **crush** fragile objects like detonators or sensitive explosives.

Impact: Mishandling could trigger a premature detonation.

Possible Solution: Implement **force sensors** or **adaptive soft grippers** to handle delicate materials.

Dependence on Color Recognition Accuracy

Challenge: Color sensors may fail in low light, harsh weather, or if the bomb is camouflaged.

Impact: The robot might not differentiate objects correctly, leading to potential **misidentification of wires** or components.

Possible Solution: Use **multi-sensor fusion** (e.g., combining **infrared**, **AI-based image recognition**, and **LiDAR**).



Applications

1. Military and Defense Operations

- Detecting and disarming explosive devices in conflict zones.
- Identifying and classifying different types of explosives based on their color-coded components.
- Safely investigating suspicious packages without risking human lives.

2. Law Enforcement and Anti-Terrorism

- Assisting bomb squads in locating, analyzing, and neutralizing potential threats.
- Recognizing hazardous materials or specific wiring colors to prevent detonation.
- Handling dangerous situations like hostage crises or public area threats.

Applications

3. Airport and Transportation Security

- Inspecting suspicious luggage or cargo for explosives using color recognition.
- Enhancing security checks in airports, train stations, and ports.
- Quickly differentiating between harmless items and potential threats.

4. Industrial and Commercial Safety

- Monitoring and neutralizing explosive risks in chemical plants or factories.
- Identifying and disposing of malfunctioning or discolored hazardous materials.
- Supporting hazardous material handling in nuclear and thermal power plants.

Applications

5. Search and Rescue Operations

- Identifying dangerous objects in disaster zones (collapsed buildings, earthquake sites).
- Safely clearing debris with potential explosive risks.
- Assisting in mine detection and clearance missions.

6. Border and Customs Control

- Examining suspicious cargo and vehicles at border checkpoints.
- Detecting contraband or hazardous materials using color-coded identification.

Applications

7. Public Event Security

- Ensuring safety at large gatherings like sports events and political rallies.
- Rapidly scanning and identifying suspicious objects in crowded areas.
- Minimizing risks by isolating and analyzing threats before they cause harm.

8. Education and Research

- Training bomb disposal squads using simulated threats.
- Advancing robotics and AI research in hazardous environments.
- Developing new techniques for remote sensing and object classification.

Contribution to Society

1. Enhanced Public Safety

-Bomb Neutralization: The robot can safely identify and neutralize explosive devices, minimizing the risk to human life.

-Crowd Protection: It can be deployed in public spaces to handle suspicious objects, ensuring the safety of large gatherings.

2. Reduced Risk to Human Life

-Remote Operation: Human bomb disposal experts can control the robot from a safe distance, reducing their exposure to danger.

-Hazardous Environments: The robot can be used in environments that are too dangerous for humans, such as chemical spills or biohazard areas.

Contribution to Society

3. Improved Efficiency and Accuracy

-Quick Decision-Making: The robot can rapidly analyze and distinguish objects, enabling faster and more accurate responses.

4. Support for Law Enforcement and Military

-Evidence Collection: The robot can safely collect evidence from a scene without contaminating or tampering potential clues.

Progress

- The assembly of the chassis for the bomb disposal robot began as follows:
- We began by selecting a durable base.
- This base is capable of supporting the robotic arm and the onboard electronics.
- A four-wheel drive chassis was chosen to ensure stability and maneuverability across various terrains.
- Essential components include:
 - ✓ DC gear motors,
 - ✓ wheels
 - ✓ L298N motor driver
- All these were procured from local electronics markets.

Technical Difficulties

- We encountered problems related to the colour recognition as we realized it wasn't a very reliable and dependable option to recognize different wires based on their functionality.

On testing the statistics we came up with a more reliable as well as technically sound and simpler solution.

Technical Difficulties:Solution

- **The current solution came from our list future scope developments . It is as follows:**

Explosive Wire Type Detection

Incorporate **resistance or capacitance measurement circuits** to try identifying which wire is live.

Can help when color isn't a reliable indicator.



Example ML Use Case for Wire Detection

Feature 1	Feature 2	Feature 3	Output
5.0 Ω	10ms	8 μ T	Live Wire
50k Ω	0ms	0 μ T	Dummy Wire
15 Ω	5ms	3 μ T	Neutral

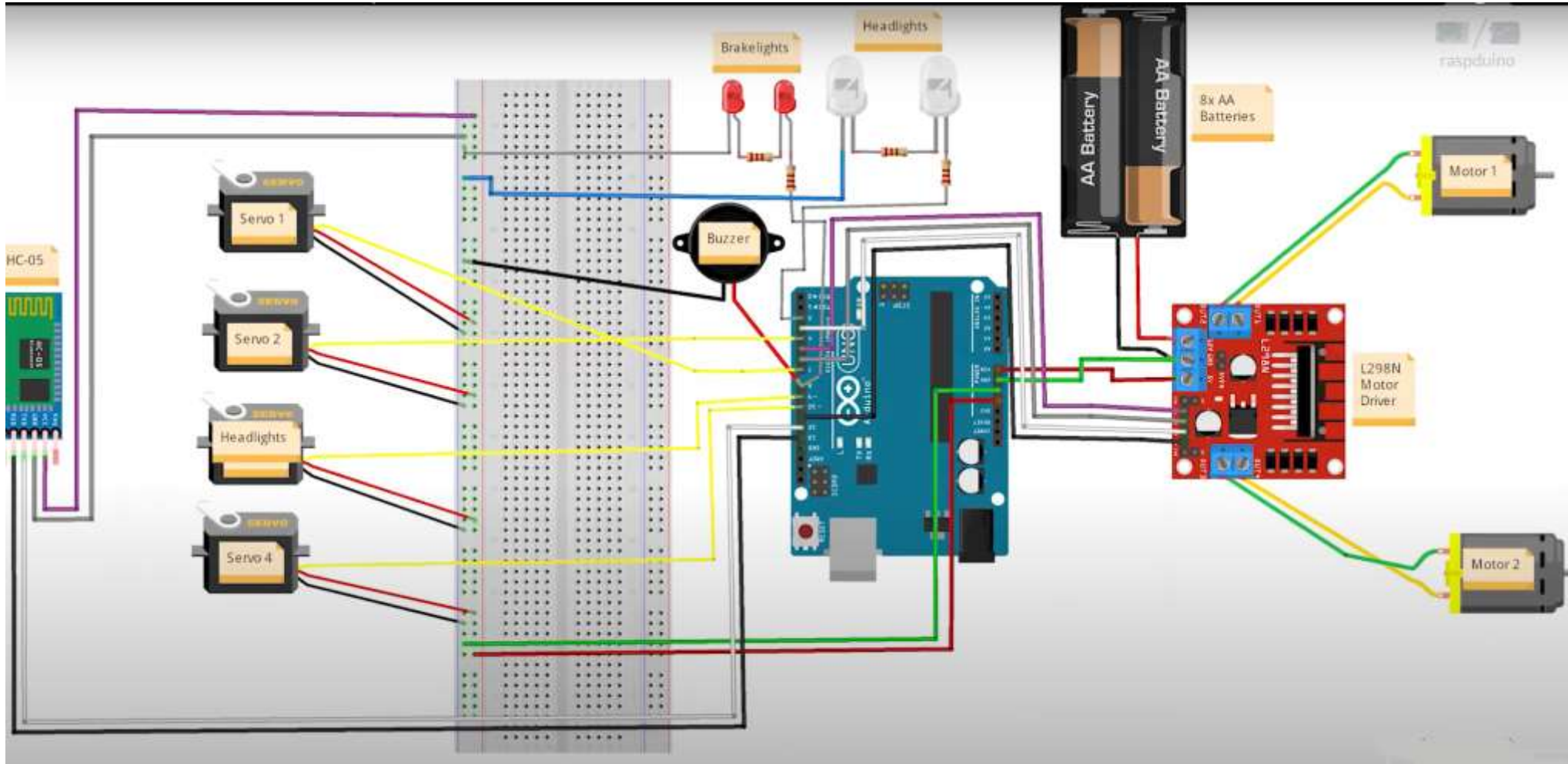
Features:

- Resistance (Ω)
- Capacitance charge time (ms)
- Magnetic field strength (μ T)

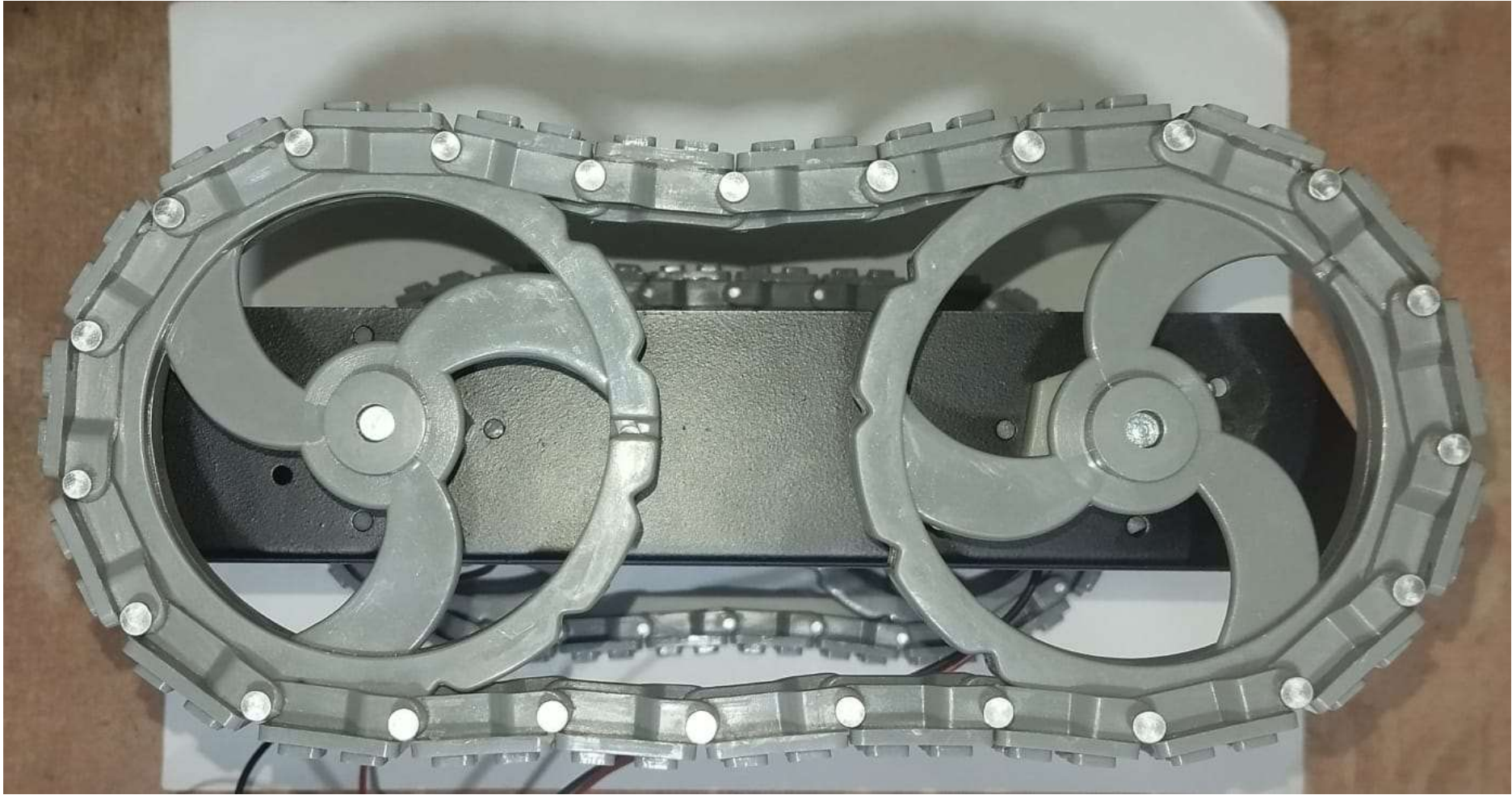
Detection Strategy:

- Move the sensor close to each wire (without touching).
- Look for **spikes or consistent changes** in magnetic field readings.
- The wire with higher or unusual magnetic activity is probably carrying current.

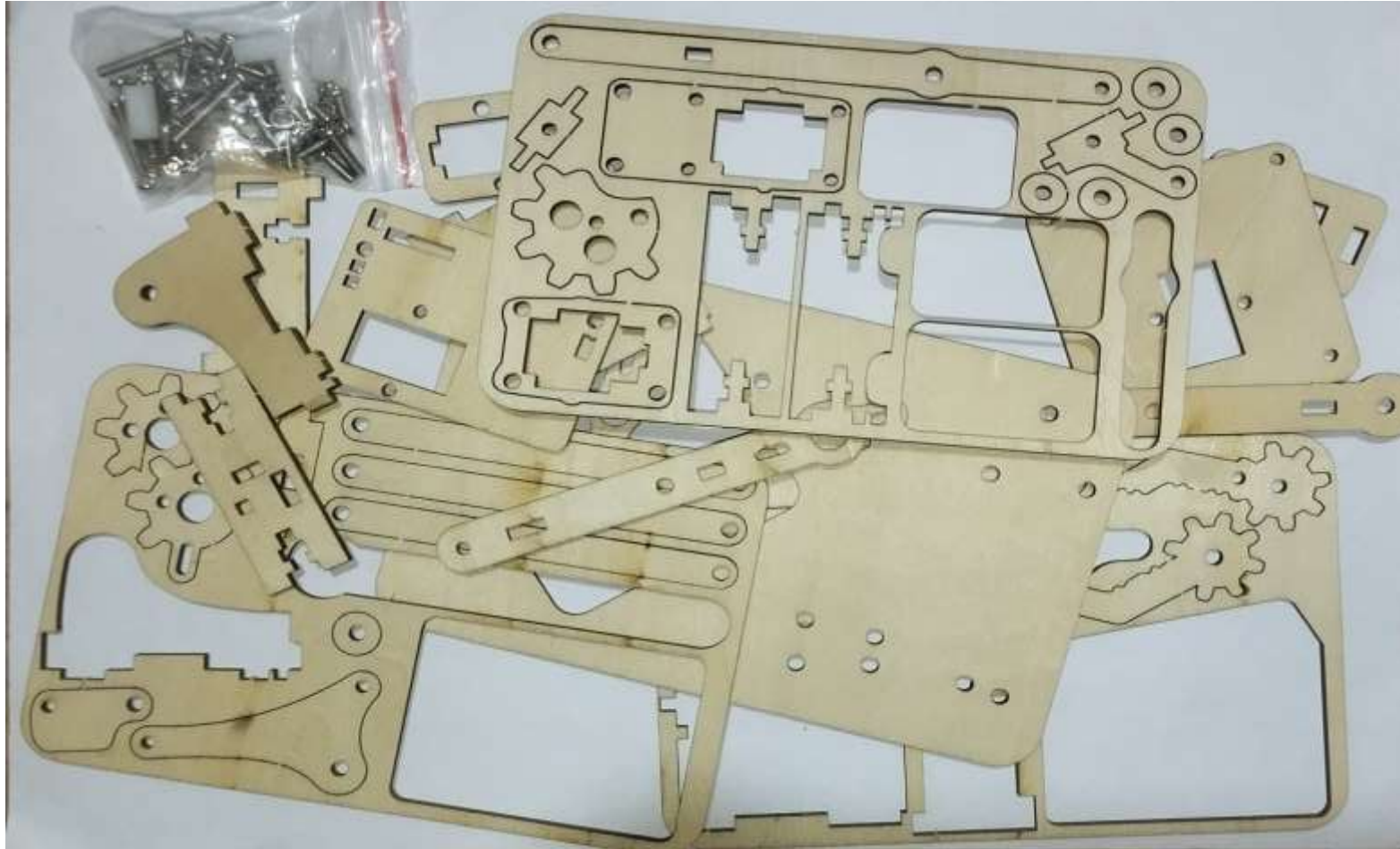
Circuit Diagram



Working on Project



Working on Project



Working on Project



Working on Project



Future Scope

Add AI-Powered Vision Upgrade TCS3200 to a **Camera Module** (e.g., OpenMV Cam, Raspberry Pi Camera). Use **machine learning** to identify not just wire colors but object types (e.g., bombs, tools, humans). **YOLO or MobileNet** models can run on lightweight platforms for object detection.

Semi-Autonomous Navigation

Add **ultrasonic or LiDAR sensors** for obstacle avoidance.

Use **line following** or **waypoint navigation** for basic autonomous movement to the target.

Implement **SLAM (Simultaneous Localization and Mapping)** with Raspberry Pi + LiDAR for advanced navigation.

Future Scope

Explosive Wire Type Detection

Incorporate **resistance or capacitance measurement circuits** to try identifying which wire is live.
Can help when color isn't a reliable indicator.

Tactical Deployment Features

Design the robot to be **foldable or compact**, deployable from a vehicle or a backpack.
Add a **quick-start mode** for emergency scenarios.

Thermal Imaging

Use an **infrared thermal camera** to detect heat signatures—useful for identifying recently handled objects, active circuits, or humans nearby.

Future Scope

Voice-Controlled Commands

Use an **offline voice recognition module** (like Arduino + ML models) for commands like “move forward,” “pick,” “cut wire,” etc.

Helps when the operator’s hands are busy or during emergencies.

Self-Diagnosis and Health Monitoring

Monitor **motor temperature, current draw, and sensor health**.

Send status reports back to the operator to warn of potential hardware failures before they happen.

References

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